

Course: BTech

Semester: 6

Prerequisite: Parallel Computing, Computer Architecture | 303105102 - Programming for Problem Solving

Rationale : HPC is more than just for achieving high performance - it is a compelling vision for how computation can seamlessly scale from a single processor to virtually limitless computing power. The market demands general-purpose processors that deliver high single threaded performance as well as multi-core throughput for a wide variety of workloads on client, server, and high- performance computing (HPC) systems

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
0	0	2	0	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Outcome

After Learning the Course the students shall be able to:

1. To understand the basic parallel programming concepts & GPU.
2. To comprehend the various High Performance Computing Paradigms.
3. To design parallel algorithms.
4. To demonstrate the Message passing and synchronization concepts.
5. To Perform basic CUDA Programming.

List of Practical

1.	Study the facilities provided by Google Colab.
2.	Demonstrate basic Linux Commands.
3.	Using Divide and Conquer Strategies design a class for Concurrent Quick Sort using C++.
4.	Demonstrate MPI functions through a simple program.
5.	Write a program to check task distribution using Gprof.
6.	Write a simple CUDA program to print "Hello World!"
7.	Write a simple CUDA program to add two numbers.
8.	Write a CUDA program to add two arrays.
9.	Analyze the code using Nvidia-Profilers.
10.	Demonstration of OpenMP and pthread functions.